

USC ADVANCE Study — Threshold Paper: Analysis Results

Table 1a. Predictors of E-cigarette Use Subgroups — Users Only (USC ADVANCE Study)

Variable	No Exposure (n=510)	No Perceived Friends (n=226)	Neither (n=142)	Low Threshold (n=610)
Demographics				
Cohort C2	1.75	0.99	1.92	0.91
Female	0.73	1.13	1.04	0.89
Hispanic/Latine	2.06	0.91	1.81	1.70
Asian	2.49	2.31	3.92*	3.53**
Sexual Minority	1.20	1.35	2.07	1.31
Parent Education	0.90	0.79	0.89	0.92
GAD	1.27	1.51	1.81	1.04
MDD	1.07	0.53	0.46*	0.79
Grade Period Fixed Effects (ref = GP3)				
GP4	1.51	—	0.72	0.81
GP5	1.73	1.99	2.05	0.68
GP6	1.52	2.56	2.38	0.56
GP7	0.87	1.91	1.07	0.28
GP8	0.75	1.99	1.04	0.26
Network Characteristics				
Out-degree (GP3)	0.81*	1.10	0.94	0.98
In-degree (GP3)	0.93	0.99	0.97	0.96
Friends (GP3)	0.74*	0.81	0.51**	0.86
Friends (TOA)	0.72**	—	—	0.78*
Friends (GP8)	0.85	0.56**	0.57**	0.85
Network Exposure (lag)	—	0.12	—	—
Network Exposure (GP8)	—	0.99	—	—
Social Media				
sm_ecig_any (W4–W8)	2.31*	0.40*	0.87	1.39

*Note. Values are ORs from logistic regression. * $p < .05$; ** $p < .01$ (bold). All models include grade period fixed effects (factor(TOA), ref = GP3) and cohort dummy. No school fixed effects (separation problems in small subsets). Subgroup definitions at TOA — No Exposure: network exposure = 0; No Perceived Friends: perceived friend e-cig use = 0; Neither: both = 0; Low Threshold: $\leq 25\%$ of network friends using (justified by threshold distribution: 70th percentile = 0.25 in this sample, consistent with Valente, 1996). — = excluded (variable directly defines subgroup or all values zero). Friends (TOA) excluded from Cols 2–3 (definitional overlap). Network Exposure rows excluded from Cols 1, 3, 4 (definitional overlap or separation). sm_ecig_any available W4–W8 only; GP3 adopters excluded from social media models.*

Table 1b. Predictors of Non-Use Subgroups — Resistant to Peer Influence (USC ADVANCE Study)

Variable	High Exposure (n=1,607)	High Perceived Friends (n=1,257)	Both High (n=793)
Demographics			
Cohort C2	0.79*	0.75**	0.87
Female	1.03	1.30**	1.63**
Hispanic/Latine	1.01	1.10	1.03
Asian	0.51**	0.34**	0.37**
Sexual Minority	0.78*	0.83	0.79
Parent Education	0.96	0.93**	0.98
GAD	0.92	1.08	0.93
MDD	1.20	1.51**	1.53**
Network Characteristics			

Out-degree (GP3)	1.39**	1.06*	1.25**
In-degree (GP3)	1.02	1.01	1.02
Friends (GP3)	1.13	—	2.03**
Friends (GP8)	1.47**	—	3.42**
Social Media			
sm_ecig_any (W4–W8)	—	—	—

Note. Values are ORs from logistic regression. * $p < .05$; ** $p < .01$ (bold). Non-user subgroups — High Exposure: max network exposure > 0 ($n=1,607$); High Perceived Friends: max perceived friend use > 0 ($n=1,257$); Both High: both conditions met ($n=793$). Friends (GP3/GP8) excluded from Col5 (definitional overlap). sm_ecig_any not available for non-users (no time of adoption); shown as — for all columns.

Table 2. Predictors of E-cigarette Adoption and Threshold — Main Model (GAD + MDD)

	Adoption N=3,530 (841 adopters, 2,689 non-adopters) 14,666 person-periods OR	Threshold (N=406) Coefficient
Constant	0.02**	-20.63
Grade Period (GP3 = reference)		
GP4	3.85**	17.23
GP5	24.38**	17.88
GP6	45.91**	18.11
GP7	74.56**	18.46
GP8	—	18.84
Cohort (C1 = reference)		
C2	1.19	0.97*
Attributes		
Female	1.12	0.26
Hispanic/Latine	0.85	0.07
Asian	0.68	-0.51
Parent Education	0.71	0.03
School Effects		
Student random intercept (1 id)	included	—
School random intercept (1 school)	included	—
School fixed effects (factor(school))	—	included
Mental Health		
GAD	0.31**	-0.03
MDD	3.44**	-0.21
Network Covariates		
Network exposure	6.46*	—
Friends who use e-cigarettes	1.64**	0.19**
Sexual minority	1.05	-0.12

Note. * $p < 0.05$; ** $p < 0.01$. Adoption: mixed-effects logistic regression with student (1|id) and school (1|school) random intercepts. $N = 3,530$ students (841 adopters, 2,689 non-adopters); 14,666 person-period observations at risk of adoption. School random intercept variance is estimated but individual school coefficients are not reported. GP8 near zero due to sparse events. Threshold: grouped binomial GLM with school fixed effects; excludes isolates ($n_alters = 0$); $N = 406$ non-isolate adopters. School 105 threshold omitted (near-perfect separation).

Table 3. Predictors of E-cigarette Adoption and Threshold — Sensitivity Model with School Fixed Effects (GAD + MDD)

	Adoption (Sensitivity) N=3,530 (841 adopters, 2,689 non-adopters)	Threshold (N=406) Coefficient
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	14,666 person-periods OR	
Constant	0.02**	-20.63
Grade Period (GP3 = reference)		
GP4	1.00	17.23
GP5	1.02	17.88
GP6	0.67*	18.11
GP7	0.56**	18.46
GP8	—	18.84
Cohort (C1 = reference)		
C2	1.10	0.97*
Attributes		
Female	1.17	0.26
Hispanic/Latine	1.14	0.07
Asian	0.77	-0.51
Parent Education	0.95	0.03
School Fixed Effects (ref = School 101)		
School 102	0.60	0.16
School 103	0.76	0.03
School 104	1.17	1.09*
School 105	1.10	—
School 106	0.81	0.69
School 107	1.28	0.89
School 112	1.22	0.84
School 113	1.05	0.89
School 114	1.49	1.29**
Mental Health		
GAD	0.77*	-0.03
MDD	1.61**	-0.21
Network Covariates		
Network exposure	2.31**	—
Friends who use e-cigarettes	1.43**	0.19**
Sexual minority	1.18	-0.12

Note. * $p < 0.05$; ** $p < 0.01$. Adoption: standard logistic regression with school fixed effects (factor(school), ref = School 101) and grade period fixed effects; $N = 14,666$ person-period observations. Results are consistent with the main mixed model (Table 2) for key predictors. School 105 threshold coefficient omitted (near-perfect separation). This table serves as a sensitivity check allowing inspection of individual school effects.